NYC Airbnb Market Analysis

Pricing, Demand & Value Optimization

Sector: Hospitality & Travel Analytics

Team ID: Team 18

Institute: Newton School of Technology

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2. Executive Summary

The Airbnb ecosystem in New York City is highly competitive, with thousands of listings varying in pricing, availability, location, and guest engagement. Travelers often struggle to identify listings that offer the best balance between affordability, comfort, and reliability.

This project analyzes the NYC Airbnb 2019 dataset to enable smarter booking decisions. Using exploratory data analysis, statistical testing, regression modeling, and clustering, key patterns in pricing and demand were identified.

The analysis reveals that pricing is primarily driven by room type and location, while demand is concentrated in budget-friendly listings. Manhattan listings command premium prices but often offer lower value-for-money, whereas Brooklyn and Queens provide a better balance between affordability and demand.

The study introduces derived metrics such as value score, demand score, and occupancy rate to improve decision-making. Based on the findings, actionable recommendations are provided for travelers, hosts, and platform optimization.

3. Sector and Business Context

The hospitality and travel sector has been transformed by digital platforms like Airbnb, enabling peer-to-peer accommodation marketplaces. In major cities like New York, thousands of listings compete across dimensions such as price, location, availability, and reviews.

Stakeholders

- Travelers seeking affordable and reliable stays
- Hosts aiming to optimize pricing and occupancy
- Platforms focusing on improving user experience

Why This Problem Matters

- Users face decision fatigue due to too many options
- Pricing varies significantly across boroughs
- Demand patterns are not easily visible
- Better insights can improve booking satisfaction

4. Problem Statement and Objectives

Problem Statement

This project aims to analyze the Inside Airbnb NYC dataset to help users identify the most suitable Airbnb listings based on pricing, neighborhood, reviews, availability, and stay preferences, enabling better booking decisions.

Objectives

- Identify key drivers of Airbnb pricing
- Understand demand distribution across boroughs
- Evaluate value-for-money across listings
- Segment listings into meaningful categories
- Provide actionable insights for better decisions

Scope & Success Criteria

- **Scope:** NYC Airbnb dataset (2019 snapshot). Focus on pricing, demand, and availability.
- **Success Criteria:** Clear pricing drivers identified, actionable insights generated, and strong statistical validation.

5. Data Description

The dataset used is **NYC Airbnb Open Data 2019**, sourced from Inside Airbnb/Kaggle.

- **Total Rows:** 45,739
- **Total Columns:** 17
- **Granularity:** One row per listing

Features Overview

- **Key Features:** Price, Neighbourhood Group, Room Type, Availability (365 days), Number of Reviews, Reviews per Month.
- **Derived Features:** Price per Person, Occupancy Rate, Value Score, Demand Score, Price Category, Availability Category.

Data Quality Issues

- Zero-price listings (data anomalies)
- Missing reviews (~20%)
- Extreme minimum night values
- Price outliers (handled)
- Dataset skewed toward Manhattan and Brooklyn

6. Cleaning and Transformation

Steps Performed

- Handled missing values (filled logically)
- Removed duplicates
- Converted date columns
- Standardized column names
- Treated outliers using IQR

Feature Engineering

- Created `price_per_person`
- Derived `occupancy_rate`
- Built `value_score` and `demand_score`
- Categorized price and availability

Final Output

A clean dataset (`processed_AB_NYC_2019.csv`) with no missing values and consistent structure.

7. KPI Framework

Core KPIs

> **Average Price**
>
> `= Total Price ÷ Number of Listings`

> **Total Listings**
>
> `= Count of Listings`

> **Average Reviews**
>
> `= Total Reviews ÷ Total Listings`

> **Average Availability**
>
> `= Total Availability ÷ Total Listings`

> **Price per Person**
>
> `= Price ÷ Number of Guests`

Occupancy Rate

```
= [(365 − availability_365) ÷ 365] × 100
```

Advanced KPIs

Value Score

```
= reviews_per_month ÷ price_per_person
```

Demand Score

```
= reviews_per_month × [1 − (availability_365 ÷ 365)]
```

Borough Popularity Index

```
= Combination of (demand_score + reviews + listings)
```

Business Importance: These KPIs help users compare listings, identify demand trends, and select optimal locations.

8. Exploratory Analysis

Key Trends

- Price distribution is right-skewed (majority under $150)
- Entire homes dominate high-price segments
- Private rooms dominate budget segments
- Manhattan and Brooklyn dominate supply

Segment Insights

- **Manhattan:** Premium pricing
- **Brooklyn:** Balanced market
- **Queens:** Best affordability
- **Bronx/Staten Island:** Low supply

9. Statistical Analysis

Methods Used

- ANOVA
- T-test
- Correlation analysis
- Linear regression
- KMeans clustering

Results

- Room type significantly impacts price
- Manhattan prices are statistically higher
- Weak correlation between reviews and availability
- Low R^2 indicates categorical dominance
- Three distinct clusters identified

Business Interpretation: Pricing decisions depend mainly on location and room type, not numerical variables alone.

10. Dashboard Walkthrough

The Tableau dashboard enables interactive decision-making.

Key Views

- Price distribution
- Room type comparison
- Value for money by borough
- Demand score analysis
- Occupancy rate trends

Filters

- Room Type
- Availability
- Price Category
- Neighborhood

11. Key Insights

- Manhattan listings are expensive but offer lower value-for-money
- Brooklyn provides the best balance of price and demand
- Budget listings receive the highest engagement
- Entire homes dominate premium pricing
- Private rooms are ideal for budget travelers
- Luxury listings underperform in demand
- Demand is concentrated in Manhattan and Brooklyn
- Pricing depends mainly on location and room type

12. Recommendations

For Travelers

- Choose Brooklyn or Queens for better value
- Prefer private rooms for affordability

For Hosts

- Adjust pricing dynamically
- Improve availability for higher occupancy

For Platform

- Highlight value-based listings
- Improve recommendation systems

Expected Impact: Better decision-making, higher occupancy rates, and improved user satisfaction.

13. Limitations and Next Steps

Limitations

- Dataset limited to 2019
- No real-time data
- Missing review information

Future Work

- Time-series analysis
- Recommendation engine
- Real-time pricing models

14. Contribution Matrix

Role	Name	Contribution	GitHub
Project Lead	Jashvitha	Project coordination and delivery	github/jash1910
Data Lead	Krish	Dataset sourcing and understanding	github/krish-2509
ETL Lead	Akhil	Data cleaning and preprocessing	github/AkhilMishra2216
Analysis Lead	Aditya	EDA and statistical analysis	github/RajAditya7777
Visualization Lead	Jashvitha	Dashboard development	github/jash1910
Strategy Lead	Abhishek	KPI framework and insights	github/HeyBoY-ops
PPT & Quality Lead	Chaitanya	Report and documentation	github/Chait000